

Measure-Valued Stochastic Control and Matching in Spatial Forestry Markets

McKean–Vlasov Growth, Unbalanced Transport, Bayesian Learning, and Risk-Sensitive Regime Switching

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Abstract

We study a partially observed stochastic control problem for a spatial forestry system whose economically relevant state is a time-varying distribution of tree sizes, qualities, locations, log recoveries, processor specifications, and operating capacities. Standing-tree state is represented by a finite positive measure on a marked tree space. Conditional McKean–Vlasov dynamics describe size and quality evolution under common environmental noise and idiosyncratic variation, while harvest and bucking induce a finite measure of heterogeneous logs. Processor demand is represented by specification sets and grade-dependent price schedules; allocation of logs through spatially heterogeneous operators is a capacitated partial-transport problem. The dual of this matching problem yields marginal values of compatible supply, demand, and operating capacity. Because both the physical state and structural parameters are imperfectly observed, the sufficient decision state is a posterior law on a product of measure spaces. We formulate the resulting Bayes-adaptive problem, characterize the topological and convexity structure of its components, state the Wasserstein-space HJB equation for the conditional-law subproblem, and separate convex inner allocation from non-convex learning and discrete regime decisions. A reduced-order East African calibration is introduced only after the structural model is specified, and is interpreted as prior-predictive evidence rather than as a primitive assumption. The framework therefore treats biological growth, log recovery, processor compatibility, spatial transport, bargaining, observation, and regime transition as distinct operators to be identified from data.

Keywords: stochastic control; McKean–Vlasov dynamics; Wasserstein geometry; unbalanced optimal transport; Bayesian experimental design; distributionally robust optimization; forestry.

Suggested arXiv categories: math.OC, math.PR, stat.AP. Throughout, $\mathbb{R}_+ := [0, \infty)$.

1 Introduction

A forestry market is a coupled biological and spatial allocation system. Standing trees evolve stochastically; harvesting transforms trees into logs; logs are heterogeneous in diameter, length, species, moisture, and quality; processors accept different subsets of that log space and attach different prices to different grades. Transport and operating costs depend on location and contractor productivity. The relevant market object is therefore a distribution of feasible material, not a scalar inventory of hectares or cubic metres.

The mathematical problem has three nested layers. The first is a *state-estimation problem*: infer the distribution of future tree and log attributes from incomplete inventory, remote sensing,

management records, and operational observations. The second is a *matching problem*: allocate heterogeneous supply to processor-specific demand subject to spatial costs and quality constraints. The third is a *sequential control problem*: choose sensing, matching exposure, contract variables, and discrete regime transitions while the parameters governing growth, recovery, acceptance, response, and operating cost are unknown.

The analysis is deliberately agnostic about a particular institutional or commercial implementation until section 8. The objective is to identify the smallest mathematical structure that preserves the features which matter for optimization. The resulting model combines four familiar objects: controlled McKean–Vlasov dynamics on Wasserstein space, finite-measure geometry for birth/death/harvest, convex optimal transport for market matching, and Bayesian control on a belief space. Dynamic programming for McKean–Vlasov control on probability-measure spaces is developed in Pham and Wei (2016); Djete et al. (2019); unbalanced optimal transport between arbitrary finite positive measures is developed in Chizat et al. (2015); Liero et al. (2015); and the control/learning coupling is the classical dual-control phenomenon, reviewed recently by Meijer and Rantzer (2026) and studied in active-learning form by Li et al. (2023).

The main contributions are analytical. First, we make the primitive state spaces and their topology explicit, separating probability-law geometry from finite-mass geometry. Second, we connect stochastic tree growth to processor economics through a harvest/bucking kernel and specification-dependent log prices. Third, we formulate processor allocation as a convex unbalanced transport problem and use its dual to define marginal supply value. Fourth, we formulate partial observation over both physical state and structural parameters, rather than treating the physical asset map as known. Fifth, we isolate which subproblems are convex and which are intrinsically non-convex. Finally, we show how the existing hectare-level model is obtained as a coarse projection of the richer state system and use East African data for a prior-predictive numerical calibration.

2 Mathematical setting and geometry

2.1 Primitive state spaces

Let (D, d_D) be a compact geographic metric space. The metric may be Euclidean at coarse resolution or a road-network/travel metric when delivered cost is evaluated. Let G be a finite set of species/genetic classes, $Q_T \subset \mathbb{R}^{q_T}$ and $Q_L \subset \mathbb{R}^{q_L}$ compact tree- and log-quality state spaces, and $M \subset \mathbb{R}^{q_M}$ a compact management-state space. For positivity of diameter and height, write tree size in logarithmic coordinates $s = (s_d, s_h) = (\log d, \log h) \in \mathbb{R}^2$. A marked tree state is

$$\xi = (\ell, a, s, g, q, m) \in E_T := D \times \mathbb{R}_+ \times \mathbb{R}^2 \times G \times Q_T \times M, \quad (1)$$

where ℓ is location, a is age, g is species/genetics, q contains physical quality/form attributes (straightness, defect or density-related states when observed), and m is management condition. Epistemic confidence is represented by the observation model and posterior, rather than embedded in the physical state.

A harvested log is represented by

$$\lambda = (\ell, d_s, d_\ell, L_L, g, q_L, w) \in E_L := D \times \mathbb{R}_+^3 \times G \times Q_L \times [0, 1], \quad (2)$$

where d_s and d_ℓ are small- and large-end diameters, L_L is log length, q_L is a vector of grade-relevant quality attributes, and w is moisture fraction or another density-relevant condition. A reference

diameter can be recovered as a measurable function of (d_s, d_ℓ, L_L) . Let $m_L(\lambda) \geq 0$ denote log mass in tonnes, with strict positivity on physically admissible logs. A generic representation is

$$m_L(\lambda) = \varrho(g, q_L, w) V_L(d_s, d_\ell, L_L, q_L), \quad (3)$$

with ϱ the relevant density map and V_L a taper/volume map. This avoids a universal cubic-metre-to-tonne conversion and makes the tree-to-log map compatible with processor specifications stated in dimensions, quality and mass.

Let $\mathbf{C}_P$ be a compact metric specification-parameter space. A point $\sigma_j^{\text{spec}} \in \mathbf{C}_P$ encodes a finite number of grades, diameter/length thresholds, accepted species/quality classes, and the corresponding grade/quantity value schedule. Processor j has state

$$y_j = (\ell_j, Q_j, \sigma_j^{\text{spec}}, v_j, R_j^{\text{src}}) \in \mathbf{E}_P := \mathbf{D} \times \mathbb{R}_+ \times \mathbf{C}_P \times [0, 1] \times \mathbb{R}_+, \quad (4)$$

where Q_j is desired procurement flow after normalization to tonnes per unit time, v_j is utilisation, and R_j^{src} is an economically tolerable sourcing radius. If a source reports demand in cubic metres, the normalization is state-dependent through (3); no universal volume-to-mass coefficient is assumed. The specification family $\mathcal{S}_j$ and price schedule $\mathbf{p}_j$ used below are measurable functions of σ_j^{spec} .

Operator k has state

$$z_k = (\ell_k, K_k, \vartheta_k, \varphi_k, c_k, \eta_k^{\text{comp}}) \in \mathbf{E}_O := \mathbf{D} \times \mathbb{R}_+^4 \times [0, 1], \quad (5)$$

where K_k is capacity, ϑ_k productivity, φ_k equipment/fuel intensity, c_k operating cost, and η_k^{comp} a compliance/quality mark.

We assume $\mathbf{E}_T, \mathbf{E}_L, \mathbf{E}_P, \mathbf{E}_O$ are Polish under their product metrics. Discrete species and grade marks are endowed with the discrete topology.

2.2 Stratification by biological marks

The discrete biological marks should not be silently treated as Euclidean coordinates. For each $g \in \mathbf{G}$, define the continuous tree and log strata

$$\mathbf{E}_T^g := \mathbf{D} \times \mathbb{R}_+ \times \mathbb{R}^2 \times \mathbf{Q}_T \times \mathbf{M}, \quad \mathbf{E}_L^g := \mathbf{D} \times \mathbb{R}_+^3 \times \mathbf{Q}_L \times [0, 1], \quad (6)$$

and identify $\mathbf{E}_T = \bigsqcup_{g \in \mathbf{G}} \mathbf{E}_T^g$ and $\mathbf{E}_L = \bigsqcup_{g \in \mathbf{G}} \mathbf{E}_L^g$. A finite disjoint union of Polish spaces is Polish. It is generally *not* geodesic across distinct species strata under the discrete mark metric, even when each continuous stratum is geodesic. Consequently, species change is not modeled as transport through the state space.

For measure geometry it is therefore useful to identify a standing-tree measure with the vector $\mu = (\mu^g)_{g \in \mathbf{G}}$, $\mu^g \in \mathcal{M}_+(\mathbf{E}_T^g)$, and use the product metric

$$d_{\text{HK}, \mathbf{G}}(\mu, \eta)^2 := \sum_{g \in \mathbf{G}} \text{HK}(\mu^g, \eta^g)^2. \quad (7)$$

The same construction applies to logs. This separates continuous growth and transport within a biological class from discrete transitions that must be introduced explicitly through a mutation/reclassification kernel if they are scientifically meaningful. The resulting state space is best viewed as a finite collection of continuous metric strata coupled by discrete kernels, rather than as a single smooth manifold.

2.3 Finite measures, Wasserstein geometry, and changing mass

Let $\mathcal{M}_+(\mathbf{E})$ denote the cone of finite nonnegative Radon measures on a Polish space $\mathbf{E}$. Fixing a base point $x_0 \in \mathbf{E}$, write

$$\mathcal{M}_{2,+}(\mathbf{E}) := \left\{ \mu \in \mathcal{M}_+(\mathbf{E}) : \int d_{\mathbf{E}}(x, x_0)^2 \mu(dx) < \infty \right\}. \quad (8)$$

The moment condition is independent of the chosen base point. Let n_P and n_O denote the numbers of active processors and operators. The standing-tree, harvested-log, processor, and operator states are respectively

$$\mu_t \in \mathcal{M}_+(\mathbf{E}_T), \quad \Lambda_t \in \mathcal{M}_+(\mathbf{E}_L), \quad \nu_t = \sum_{j=1}^{n_P} \delta_{y_j(t)}, \quad \kappa_t = \sum_{k=1}^{n_O} \delta_{z_k(t)}. \quad (9)$$

The total mass of μ_t is an intensity/count (or an area-weighted count under a chosen sampling convention), not a probability. Whenever $\mu_t(\mathbf{E}_T) > 0$, define its normalized law by

$$\bar{\mu}_t := \frac{\mu_t}{\mu_t(\mathbf{E}_T)} \in \mathcal{P}_2(\mathbf{E}_T), \quad (10)$$

provided the second moment is finite.

Two geometries are required and should not be conflated. The quadratic Wasserstein metric W_2 acts on $\mathcal{P}_2(\mathbf{E}_T)$ and is natural for the conditional distribution of a representative surviving tree. Harvesting, mortality, recruitment, and unmatched market supply alter total mass, so arbitrary finite measures are instead endowed with a mass-varying metric. We use the Hellinger–Kantorovich metric HK, equivalently an entropy-transport/unbalanced-transport geometry, which combines displacement with creation or destruction of mass (Chizat et al., 2015; Liero et al., 2015). On geodesic ground strata these metric spaces admit non-Euclidean geodesics; HK also has a cone representation and a dynamic transport–reaction formulation. Its geodesics and geodesic convexity have been characterized in substantially greater detail in Liero et al. (2022); Gallouët et al. (2021). Entropic formulations admit scalable Sinkhorn-type algorithms (Chizat et al., 2016). No global geodesic property is assumed when the geographic metric or discrete mark structure fails to be geodesic.

Assume each ground space is complete and separable under the stated product metric, and equip each finite-measure component with the standard HK topology. Let $\mathbf{R}$ be a finite set of observation/coverage regimes and $\mathbf{Z} \subset \mathbb{R}^d$ a Polish finite-dimensional state. Define

$$\mathbf{X} = \mathcal{M}_+(\mathbf{E}_T) \times \mathcal{M}_+(\mathbf{E}_P) \times \mathcal{M}_+(\mathbf{E}_O) \times \mathbf{R} \times \mathbf{Z}. \quad (11)$$

The harvested-log measure Λ_t is treated here as an endogenous flow generated from the persistent standing-tree state by the harvest/bucking operator in section 3; it can be appended as an additional $\mathcal{M}_+(\mathbf{E}_L)$ coordinate without changing the product-space arguments below when inventory-in-process must itself be persistent.

Proposition 1 (Polish product and belief spaces). *Under the preceding assumptions, $\mathbf{X}$ is Polish under any compatible product metric. If $\Theta \subset \mathbb{R}^p$ is closed and $d_{\mathbf{X}}$ is such a metric, then*

$$\mathcal{P}_2(\mathbf{X} \times \Theta) := \left\{ \pi : \int (d_{\mathbf{X}}(x, x_0)^2 + \|\theta\|^2) \pi(dx, d\theta) < \infty \right\} \quad (12)$$

is Polish under the quadratic Wasserstein metric induced by the product ground metric.

Proof. Finite products of Polish spaces are Polish. The second assertion is the standard completeness and separability result for $\mathcal{P}_2$ over a Polish ground space, restricted to laws with finite second moment. $\square$

There are also two distinct convexity notions. First, $\mathcal{M}_+(\mathbf{E})$, and hence $\mathcal{M}_{2,+}(\mathbf{E})$, is a convex cone under linear interpolation $(1 - \vartheta)\mu_0 + \vartheta\mu_1$. Second, whenever the relevant ground stratum is geodesic, $\mathcal{P}_2(\mathbf{E})$ and the corresponding finite-measure space admit non-Euclidean geodesics under W_2 and HK, respectively. Linear convexity and geodesic (displacement) convexity are not equivalent. This distinction matters for existence, regularity, and algorithms on measure spaces: linear mixtures are natural for relaxed controls and resource measures, whereas Wasserstein/HK geodesics are natural for transport and gradient-flow structure (Bolley and Carrillo, 2013; Liero et al., 2015).

Definition 1 (Metric geodesic convexity). *Let $(\mathcal{Y}, d)$ be a geodesic metric space. A functional $\mathcal{F} : \mathcal{Y} \rightarrow (-\infty, \infty]$ is λ -geodesically convex if for every pair y_0, y_1 there exists a constant-speed geodesic $(y_s)_{s \in [0,1]}$ such that*

$$\mathcal{F}(y_s) \leq (1 - s)\mathcal{F}(y_0) + s\mathcal{F}(y_1) - \frac{\lambda}{2}s(1 - s)d(y_0, y_1)^2. \quad (13)$$

For $\lambda > 0$, geodesic convexity gives uniqueness of a minimizer along the admissible geodesic class whenever a minimizer exists; lower semicontinuity and compact/coercive sublevel conditions provide the corresponding direct-method existence route. In the present system, this structure can be exploited for smooth transport–reaction or regularized inference functionals, but it is not claimed for hard processor thresholds, discrete harvest rules, or impulse decisions. The distinction between linear and metric convexity is therefore part of the model specification rather than a stylistic choice.

For a fixed discrete regime and biological stratum, the continuous part of the state inherits a length-space geometry whenever the ground metric spaces do. The global state remains stratified because regime labels and species marks are discrete. Thus a single global Hessian or geodesic-convexity claim would be inappropriate: second-order analysis is performed on continuous strata, while discrete transitions are handled by impulse or switching operators. This stratified viewpoint is also useful numerically because the convex transport subproblem can be solved separately on each species/grade stratum and then coupled through processor and operator capacity constraints.

2.4 Tangent geometry and mass-changing continuity equations

On the continuous coordinates of $\mathcal{P}_2(\mathbb{R}^d)$, the formal Wasserstein tangent space at an absolutely continuous law μ is

$$T_\mu \mathcal{P}_2(\mathbb{R}^d) = \overline{\{\nabla \varphi : \varphi \in C_c^\infty(\mathbb{R}^d)\}}^{L^2(\mu)}. \quad (14)$$

An absolutely continuous curve (μ_t) admits a velocity field $v_t \in T_{\mu_t} \mathcal{P}_2$ satisfying the continuity equation

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0 \quad (15)$$

in the weak sense, with metric speed determined by the minimum $L^2(\mu_t)$ norm of admissible velocity fields. This is the geometric basis for Wasserstein gradients and Lions derivatives in the HJB layer below.

For the finite-mass forestry state, pure transport is insufficient. A generalized continuity equation has the form

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = \alpha_t^{\text{rxn}} \mu_t + \mathcal{B}_t, \quad (16)$$

where α_t^{rxn} is a signed reaction rate (negative for mortality/harvest removal) and $\mathcal{B}_t$ is a nonnegative source-rate measure representing recruitment/planting. Hellinger–Kantorovich geometry metrizes precisely this combination of transport and mass change through an action involving both displacement and reaction terms, up to the chosen normalization convention (Chizat et al., 2015; Liero et al., 2015). In the forestry interpretation, growth moves mass through size–quality space, while mortality, harvest, and recruitment change mass. Thus the geometry is not decorative: it matches the decomposition of the physical dynamics.

Geodesic convexity is not assumed globally. Processor grade thresholds, discrete species marks, harvest impulses, and regime impulses generate nonsmooth or non-convex structure. Where a functional is geodesically convex, Wasserstein/HK first-order methods can exploit that structure; elsewhere the model relies on convexification, decomposition, or discrete optimization.

2.5 Control topology and convexification

Let $\mathbf{A}_G$ be a compact metric biological-action space and let $\mathbf{U}_c$ be a compact metric space of continuous information/market actions. The full decision space is $\mathbf{U} = \mathbf{U}_c \times \mathbf{I}$, where $\mathbf{I}$ is a finite impulse set for discrete regime changes. More generally, for any compact metric action space $\mathbf{A}$, a strict control is a progressively measurable $\mathbf{A}$ -valued process. Following the standard control-space construction used in conditional McKean–Vlasov problems, one may metrize strict controls by

$$\Delta(\alpha, \beta) := \mathbb{E} \left[\int_0^T d_{\mathbf{A}}(\alpha_t, \beta_t) dt \right]. \quad (17)$$

A related internal theoretical note uses this Krylov-type metric with a Polish control space and Wasserstein-Lipschitz state coefficients (Misoka Ltd., 2025). When the strict action set is non-convex, relaxed controls take values in $\mathcal{P}(\mathbf{A})$. Since $\mathbf{A}$ is compact, $\mathcal{P}(\mathbf{A})$ is compact under weak convergence and is convex. If the local drift and cost are affine in the relaxed control, this convexification is especially useful for existence and numerical optimization.

For orientation, the convexity structure of the model is summarized in table 1.

Table 1: Convex and non-convex components of the model.

Object	Structure	Consequence
Finite positive measures	Convex cone; HK metric, geodesic on suitable ground strata	Supports mixtures and mass-changing transport
Relaxed control set $\mathcal{P}(\mathbf{A})$	Convex, weakly compact for compact $\mathbf{A}$	Useful existence/relaxation device
Log–processor–operator allocation set	Convex	Inner allocation is a continuous linear/convex program
CVaR epigraph	Convex for convex loss	Tractable risk constraints in scenario programs
Grade thresholds / discrete regime	Mixed discrete-continuous	Outer problem is generally non-convex
Endogenous adoption and network feedback	Generally non-convex	Requires global, dynamic, or decomposition methods
Chance constraints	Generally non-convex	Approximation or distributionally robust reformulation required

The optimization architecture follows this geometry. Conditional on a biological stratum, a fixed discrete regime, and fixed processor grade labels, the allocation variable lies in a convex cone of finite measures and the capacity constraints are affine. Continuous measurement-error regularization and concave quantity-value functions therefore preserve a convex inner problem. Biological impulses, hard grade labels, endogenous participation, and regime switching are retained outside that convex core. Convexification is used only where it preserves the physical state variables, rather than by averaging away processor specifications or log quality.

3 Stochastic tree growth and the log-supply map

3.1 Conditional McKean–Vlasov growth

Fix a complete product probability space

$$(\Omega, \mathcal{F}, \mathbb{P}) = (\Omega^0 \times \Omega^1, \mathcal{F}^0 \otimes \mathcal{F}^1, \mathbb{P}^0 \otimes \mathbb{P}^1), \quad (18)$$

where Ω^0 supports the common Brownian motion W^0 and Ω^1 supports the idiosyncratic Brownian motion B , with the usual completed filtrations. For a representative surviving tree, let

$$\Xi_t = (s_{d,t}, s_{h,t}, q_t) \in \mathbb{R}^{2+q_T}, \quad da_t = dt, \quad (19)$$

where $s_d = \log d$, $s_h = \log h$, and $a_t = a_0 + t$ along a cohort. Conditional on fixed marks $\varsigma = (\ell, g, m)$ and a progressively measurable biological/management action $a_t^G \in A_G$, consider

$$\begin{aligned} d\Xi_t &= b_G(\Xi_t, \hat{\mu}_t^G, a_t, \varsigma, a_t^G) dt + \sigma_G(\Xi_t, \hat{\mu}_t^G, a_t, \varsigma, a_t^G) dB_t \\ &\quad + \sigma_G^0(\Xi_t, \hat{\mu}_t^G, a_t, \varsigma, a_t^G) dW_t^0, \end{aligned} \quad (20)$$

with conditional law

$$\hat{\mu}_t^G := \mathcal{L}(\Xi_t \mid \mathcal{F}_t^0) \in \mathcal{P}_2(\mathbb{R}^{2+q_T}). \quad (21)$$

For a fixed mark class ς , $\hat{\mu}_t^G$ is the size–quality conditional law; the normalized full marked-tree law $\bar{\mu}_t$ in (10) is obtained by mixing these conditional laws over age, location, species/genetics, and management marks. The action a_t^G can represent silvicultural decisions when the biological layer is optimized jointly; if such decisions are outside the controller’s scope, a_t^G is fixed or treated as an exogenous management process. The common noise W^0 can represent site-year climatic shocks, the law argument $\hat{\mu}_t^G$ can encode competition or stand-density effects, and B captures tree-level heterogeneity. Quality therefore evolves jointly with diameter and height rather than being attached ex post at harvest. The product-space/common-noise construction, a Polish control space, and Wasserstein-Lipschitz regularity are precisely the standard conditional McKean–Vlasov setting used in the accompanying theoretical note (Misoka Ltd., 2025) and in the control literature (Pham and Wei, 2016; Djete et al., 2019).

Assumption 1 (Growth regularity). *There exists $L < \infty$ such that, uniformly on finite-age intervals in (a, ς, a^G) ,*

$$\begin{aligned} &\|b_G(x, \mu, a, \varsigma, a^G) - b_G(x', \mu', a, \varsigma, a^G)\| + \|\sigma_G(x, \mu, a, \varsigma, a^G) - \sigma_G(x', \mu', a, \varsigma, a^G)\| \\ &\quad + \|\sigma_G^0(x, \mu, a, \varsigma, a^G) - \sigma_G^0(x', \mu', a, \varsigma, a^G)\| \leq L(\|x - x'\| + W_2(\mu, \mu')), \end{aligned} \quad (22)$$

and the coefficients have linear growth and are continuous in a^G .

Proposition 2 (Well-posed conditional-law dynamics). *Under the growth regularity assumption, a square-integrable initial condition, and an admissible a^G , (20) admits a unique strong solution on a finite horizon. Its conditional law flow lies in $\mathcal{P}_2(\mathbb{R}^{2+q_T})$.*

Proof sketch. Apply the standard fixed-point argument for McKean–Vlasov SDEs with Lipschitz dependence on state and conditional law under W_2 ; common noise is handled through conditioning on $\mathcal{F}_t^0$. See Pham and Wei (2016). $\square$

Log-size coordinates preserve positivity under Gaussian perturbations. Equation (20) is not intended to force a particular growth law: deterministic increment models, Gaussian-process drift surfaces, jump mortality, or size-structured branching models can replace the diffusion provided they induce a predictive law on harvestable size and quality. The market layer below depends on that induced law, not on the choice of biological parametrization itself.

3.2 Harvest, mortality, and bucking

Let $\lambda_t^H(\xi; u_t) \geq 0$ be a predictable harvest hazard/intensity under decision u_t , and let $\mathcal{B}_t \in \mathcal{M}_+(\mathbf{E}_T)$ denote a predictable recruitment/planting-rate measure. Mortality is represented by a second hazard $\lambda_t^M(\xi)$. These source and sink terms make total standing-tree mass non-conservative, which is why the finite-measure geometry of section 2 is retained alongside the normalized law used in (20).

On a fixed mark class ς , write μ_t^ς for the corresponding finite measure on size–quality state and $\mathcal{B}_t^\varsigma$ for the restricted recruitment-rate measure. Define the time-dependent diffusion generator

$$\begin{aligned} \mathfrak{L}_t^{\varsigma, a^G} \varphi(x) &:= b_G(x, \hat{\mu}_t^G, a_t, \varsigma, a^G) \cdot \nabla \varphi(x) \\ &+ \frac{1}{2} \text{Tr} \left[(\sigma_G \sigma_G^\top + \sigma_G^0 (\sigma_G^0)^\top)(x, \hat{\mu}_t^G, a_t, \varsigma, a^G) D^2 \varphi(x) \right]. \end{aligned} \quad (23)$$

For a smooth compactly supported test function φ , the conditional finite-measure dynamics then have the weak form

$$\begin{aligned} d\langle \varphi, \mu_t^\varsigma \rangle &= \left[\langle \mathfrak{L}_t^{\varsigma, a^G} \varphi, \mu_t^\varsigma \rangle - \langle (\lambda_t^H + \lambda_t^M) \varphi, \mu_t^\varsigma \rangle + \langle \varphi, \mathcal{B}_t^\varsigma \rangle \right] dt \\ &+ \left\langle \sigma_G^0(\cdot, \hat{\mu}_t^G, a_t, \varsigma, a_t^G)^\top \nabla \varphi, \mu_t^\varsigma \right\rangle \cdot dW_t^0. \end{aligned} \quad (24)$$

The drift part is a transport–diffusion–reaction equation; the final term is the common-noise stochastic transport term. Equation (24) makes explicit the link between the normalized conditional-law SDE and the mass-changing geometry in (16).

Let $K_B(d\lambda \mid \xi, \beta_t^B)$ be a bucking kernel: conditional on harvested-tree state ξ and bucking rule β_t^B , it returns a finite measure of log types. Over a short decision interval, the expected log-count intensity is

$$\Lambda_t(A) = \int_{\mathbf{E}_T} \lambda_t^H(\xi; u_t) K_B(A \mid \xi, \beta_t^B) \mu_t(d\xi), \quad A \in \mathcal{B}(\mathbf{E}_L). \quad (25)$$

The corresponding tonne-weighted measure is

$$\tilde{\Lambda}_t(d\lambda) = m_L(\lambda) \Lambda_t(d\lambda). \quad (26)$$

Thus commercially available mass is an image of the stochastic tree-size/quality measure under harvest timing, bucking, species-dependent density, moisture, and operational feasibility. A scalar hectares-to-tonnes coefficient is only a reduced-order approximation to the map $\mu_t \mapsto \tilde{\Lambda}_t$.

3.3 Controlled bucking kernel and downstream value

The bucking map need not be exogenous. Let $A_B(\xi)$ be the nonempty compact set of feasible cutting patterns for a harvested tree in state ξ , constrained by stem length, taper, defect positions and minimum piece dimensions. For a bounded upper-semicontinuous downstream log-value function $\mathbf{v}_t^L : E_L \rightarrow \mathbb{R}$, define

$$\mathcal{J}_B(\xi, \beta; \mathbf{v}_t^L) := \int_{E_L} \mathbf{v}_t^L(\lambda) K_B(d\lambda \mid \xi, \beta) - c_B(\xi, \beta), \quad (27)$$

where c_B is bucking/processing cost. A downstream-value-optimal bucking rule satisfies

$$\beta_t^{B,*}(\xi) \in \arg \max_{\beta \in A_B(\xi)} \mathcal{J}_B(\xi, \beta; \mathbf{v}_t^L). \quad (28)$$

The downstream value $\mathbf{v}_t^L$ may be taken from the dual value of the downstream matching problem, so the optimal cut pattern can depend on current processor specifications and spatial scarcity rather than on volume recovery alone.

Proposition 3 (Existence of a measurable bucking selector). *Suppose the graph of $\xi \mapsto A_B(\xi)$ is measurable with nonempty compact values and $(\xi, \beta) \mapsto \mathcal{J}_B(\xi, \beta; \mathbf{v}_t^L)$ is measurable in ξ and upper semicontinuous in β . Then the argmax in (28) is nonempty and admits a measurable selector.*

Proof sketch. Compactness and upper semicontinuity give pointwise attainment; a measurable maximum/selection theorem yields a measurable maximizer under the stated graph and measurability assumptions. $\square$

4 Processor specifications, matching, and price formation

4.1 Specification sets and grade-dependent price schedules

For processor j , let n_j^G be the number of active grades encoded by σ_j^{spec} , and let $\{\mathcal{S}_{jr}\}_{r=1}^{n_j^G}$ be Borel grade sets in log space, ordered from highest to lowest value. A representative grade set is

$$\mathcal{S}_{jr} = \{\lambda \in E_L : d_s \geq d_{jr}^{\min}, L_L \geq L_{jr}^{\min}, g \in G_{jr}, q_L \in Q_{jr}\}, \quad (29)$$

where $G_{jr} \subseteq G$ and $Q_{jr} \subseteq Q_L$ are the accepted species/genetic and quality sets. A processor may additionally impose upper diameter bounds, taper constraints involving (d_s, d_ℓ) , moisture bands, defect limits, or piece-count conditions; these are represented by further measurable inequalities in λ . When the continuous quality set is convex and the dimensional restrictions are affine, each fixed-species grade cell is convex in the continuous log coordinates. The finite union over species and grades is generally non-convex, and hard grade boundaries induce nonsmooth value functions.

To avoid ambiguity on overlaps, define the grade map by the highest-valued admissible grade,

$$\mathfrak{g}_j(\lambda) := \min\{r \in \{1, \dots, n_j^G\} : \lambda \in \mathcal{S}_{jr}\}, \quad \min \emptyset := \emptyset. \quad (30)$$

For a posted schedule quoted per tonne,

$$p_j^0(\lambda) = \begin{cases} p_{j, \mathfrak{g}_j(\lambda)}, & \mathfrak{g}_j(\lambda) \neq \emptyset, \\ 0, & \mathfrak{g}_j(\lambda) = \emptyset. \end{cases} \quad (31)$$

Here $p_{jr} \geq 0$ is the posted per-tonne value for grade r . Thus per-tonne value is a measurable function of physical dimensions, species and quality rather than a scalar attached to undifferentiated volume.

Procurement value may also depend on accepted quantity, and the dependence can differ by grade. Let $\pi_{jr}^{\text{marg}} : [0, Q_j] \rightarrow \mathbb{R}_+$ be processor j 's marginal willingness to pay for an additional tonne of grade r , and define the cumulative grade-value function

$$\Phi_{jr}(m) := \int_0^m \pi_{jr}^{\text{marg}}(z) dz, \quad 0 \leq m \leq Q_j. \quad (32)$$

The posted schedule (31) is the special case $\pi_{jr}^{\text{marg}} \equiv p_{jr}$. If π_{jr}^{marg} is nonincreasing, then Φ_{jr} is concave, so diminishing marginal procurement value can be represented without destroying convexity of the continuous allocation layer. Conversely, minimum lots, take-or-pay clauses, or increasing marginal price blocks can generate non-convex or mixed-integer structure.

Hard specifications are also statistically delicate because inventory and intake measurements are noisy. For grade r , let $a_{jr}^\eta(\lambda) \in [0, 1]$ be the continuous probability that log state λ is classified into grade r after convolution with a measurement-error kernel of scale $\eta > 0$, with $\sum_r a_{jr}^\eta(\lambda) \leq 1$. Define total compatibility $a_j^\eta(\lambda) := \sum_r a_{jr}^\eta(\lambda)$ and the measurement-error-regularized expected posted value

$$\bar{p}_j^\eta(\lambda) := \sum_{r=1}^{n_j^G} p_{jr} a_{jr}^\eta(\lambda). \quad (33)$$

Away from grade boundaries, the hard classification is recovered as $\eta \downarrow 0$ under a consistent measurement-error model. This distinction matters computationally: weak convergence of a log measure need not produce continuous changes in a hard-threshold grade count when positive mass lies on a specification boundary, whereas bounded continuous compatibility kernels do.

Proposition 4 (Continuity of smoothed compatible supply). *Let $k_{j,r} : \mathbf{E}_L \rightarrow [0, 1]$ be bounded and continuous and define*

$$S_{jr}^\eta(\Lambda) := \int k_{j,r}(\lambda) a_{jr}^\eta(\lambda) \Lambda(d\lambda), \quad (34)$$

where a_{jr}^η is bounded and continuous. If $\Lambda_n \rightarrow \Lambda$ narrowly, then $S_{jr}^\eta(\Lambda_n) \rightarrow S_{jr}^\eta(\Lambda)$. Hence the same conclusion holds for any stronger state topology under which convergence implies narrow convergence. For hard indicators, continuity holds at Λ whenever the relevant grade and distance sets are Λ -continuity sets.

Proof. The first statement follows from narrow convergence applied to the bounded continuous integrand $k_{j,r} a_{jr}^\eta$. The second is immediate by composition. The final statement is the Portmanteau theorem. $\square$

The existing computational schema represents processor demand through dimensional thresholds, species, grade and per-tonne price fields; repository records explicitly marked as test data are excluded from empirical calibration, but the schema identifies the relevant state variables (EA Forests, 2026).

Let $c_t(\lambda, j, k)$ be the non-stumpage cost per tonne of processing and delivering log type λ to processor j through operator k . It may include felling, extraction, loading, haulage, quality assurance, compliance, and risk terms:

$$c_t = c_t^{\text{fell}} + c_t^{\text{extract}} + c_t^{\text{load}} + c_t^{\text{haul}} + c_t^{\text{QA}} + c_t^{\text{comp}} + c_t^{\text{risk}}. \quad (35)$$

Define the compatible delivered surplus density

$$s_t(\lambda, j, k) = p_j^0(\lambda) - c_t(\lambda, j, k), \quad (36)$$

and set $s_t = -\infty$ (or equivalently forbid the edge) when $\log \lambda$ fails processor j 's specification or operator k is infeasible for that movement.

4.2 Capacitated partial transport through operators

Let J_t and K_t denote the active processor and operator index sets. Define demand and operator-capacity measures

$$D_t = \sum_{j \in J_t} Q_j \delta_j, \quad O_t = \sum_{k \in K_t} K_k \delta_k, \quad (37)$$

where both Q_j and K_k are tonnes per decision period. An allocation is a nonnegative measure Γ on $E_L \times J_t \times K_t$. The unregularized allocation problem is

$$\begin{aligned} \mathcal{V}_t(\tilde{\Lambda}_t, D_t, O_t) = \sup_{\Gamma \geq 0} \int s_t(\lambda, j, k) \Gamma(d\lambda, dj, dk) \\ \text{s.t. } \Gamma_L \leq \tilde{\Lambda}_t, \quad \Gamma_J \leq D_t, \quad \Gamma_K \leq O_t, \end{aligned} \quad (38)$$

where $\Gamma_L, \Gamma_J, \Gamma_K$ are the three marginals. This is a continuous linear program: unmatched logs, unsatisfied processor demand, and idle operator capacity are allowed.

To retain grade-specific quantity effects, define

$$M_{jr}^\eta(\Gamma) := \int a_{jr}^\eta(\lambda) \Gamma(d\lambda, \{j\}, dk), \quad \mathbf{M}_j^\eta(\Gamma) := (M_{j1}^\eta(\Gamma), \dots, M_{jn_j^G}^\eta(\Gamma)). \quad (39)$$

The quantity-sensitive extension is

$$\sup_{\Gamma \geq 0} \left\{ \sum_j \sum_{r=1}^{n_j^G} \Phi_{jr}(M_{jr}^\eta(\Gamma)) - \int c_t(\lambda, j, k) d\Gamma \right\} \quad \text{subject to the marginal constraints in (38).} \quad (40)$$

If every Φ_{jr} is upper semicontinuous and concave, (40) is a convex optimization problem in the standard sense of maximizing a concave functional over a convex feasible set. Strict concavity identifies the optimal expected grade-quantity vector even when the transporting allocation itself is not unique.

For computation, let $\mathfrak{R}_t$ be a finite reference measure with full support on the feasible set of triples. A smooth entropy-transport relaxation is

$$\begin{aligned} \inf_{\Gamma \geq 0} \int -s_t d\Gamma + \varepsilon_\Gamma \text{KL}(\Gamma \| \mathfrak{R}_t) \\ + \varepsilon_L \text{KL}(\Gamma_L \| \tilde{\Lambda}_t) + \varepsilon_D \text{KL}(\Gamma_J \| D_t) + \varepsilon_O \text{KL}(\Gamma_K \| O_t), \end{aligned} \quad (41)$$

with positive regularization/relaxation parameters $\varepsilon_\Gamma, \varepsilon_L, \varepsilon_D, \varepsilon_O$, and where KL denotes the generalized relative entropy for finite positive measures. This is an entropy-transport problem on finite positive measures (Chizat et al., 2015; Liero et al., 2015).

Proposition 5 (Existence and convexity of the inner allocation). *Assume $\tilde{\Lambda}_t, D_t, O_t$ are finite and the feasible relation in $E_L \times J_t \times K_t$ is closed. If s_t is bounded above and upper semicontinuous on that relation, the feasible set of (38) is convex and narrowly compact and an optimizer exists. If each Φ_{jr} in (40) is upper semicontinuous and concave, the quantity-sensitive extension also attains its optimum. The objective in (41) is convex in Γ ; if an active entropy term is strictly convex on the effective support, the regularized optimizer is unique there.*

Proof. Marginalization is linear, hence the marginal inequalities define a convex feasible set. Their finite upper bounds imply tightness; closedness of the marginal constraints and feasible relation gives narrow closedness, hence compactness by Prokhorov's theorem. Upper semicontinuity of the linear integral functional yields existence. The maps $\Gamma \mapsto M_{jr}^\eta(\Gamma)$ are linear and narrowly continuous because a_{jr}^η is bounded and continuous; composition with upper-semicontinuous concave Φ_{jr} preserves upper semicontinuity and concavity, so the direct method also applies to (40). Kullback–Leibler divergence is convex in its first argument, and composition with linear marginal maps preserves convexity; strict convexity gives uniqueness on the effective support. $\square$

Proposition 6 (Concavity and homogeneity of matching value). *For fixed surplus kernel s_t , let $S \in \mathcal{M}_+(\mathbf{E}_L)$, $D \in \mathcal{M}_+(\mathbf{J}_t)$, and $O \in \mathcal{M}_+(\mathbf{K}_t)$ be generic supply, demand, and operator-capacity resource measures. Then the value $\mathcal{V}_t(S, D, O)$ of (38) is jointly concave in (S, D, O) . Moreover, for $a \geq 0$,*

$$\mathcal{V}_t(aS, aD, aO) = a \mathcal{V}_t(S, D, O). \quad (42)$$

Proof. If Γ_1 and Γ_2 are feasible for resource triples (S_1, D_1, O_1) and (S_2, D_2, O_2) , then $\vartheta\Gamma_1 + (1-\vartheta)\Gamma_2$ is feasible for the convex combination of the resource triples and achieves the corresponding convex combination of objective values. Taking suprema proves concavity. Scaling a feasible allocation by a proves positive homogeneity in both directions. $\square$

The matched tonne flow into processor j and its demand-coverage ratio are

$$M_j(t) := \Gamma_j^*(\{j\}), \quad \chi_j(t) := \begin{cases} M_j(t)/Q_j, & Q_j > 0, \\ 0, & Q_j = 0. \end{cases} \quad (43)$$

By construction $0 \leq \chi_j \leq 1$. This is the endogenous notion of commercially useful supply: it depends simultaneously on size, quality, location, processor specification, operator capacity, and delivered cost.

4.3 Dual values, outside options, and bargaining

The linear-program dual of (38) is

$$\begin{aligned} \inf_{\phi, \psi, \omega \geq 0} \quad & \int \phi(\lambda) \tilde{\Lambda}_t(d\lambda) + \sum_j \psi_j Q_j + \sum_k \omega_k K_k \\ \text{s.t.} \quad & \phi(\lambda) + \psi_j + \omega_k \geq s_t(\lambda, j, k) \quad \text{on feasible triples.} \end{aligned} \quad (44)$$

At points of differentiability, optimal dual variables are the marginal values of additional compatible log supply, processor demand, and operator capacity; more generally they form supergradients of the concave resource-value function in the corresponding directions. Consequently, an inventory observation is valuable not merely when it discovers tonnes, but when it discovers tonnes in regions of log space with high dual value.

Transport shadow values are not negotiated stumpage prices. Fix processor j . Let the seller's best credible alternative net value be

$$o_j(\lambda) = \sup_{j' \neq j} \sup_{k \in \mathbf{K}_t} [p_{j'}^0(\lambda) - c_t(\lambda, j', k)]_+. \quad (45)$$

Let the best feasible net value with buyer j be

$$v_j(\lambda) = \sup_{k \in \mathbf{K}_t} [p_j^0(\lambda) - c_t(\lambda, j, k)]. \quad (46)$$

When $v_j(\lambda) > o_j(\lambda)$, a stylized Nash split with seller bargaining weight $\beta_{\text{barg}} \in [0, 1]$ gives a forest-gate net value

$$p_j^{\text{FG}}(\lambda) = o_j(\lambda) + \beta_{\text{barg}}(v_j(\lambda) - o_j(\lambda)). \quad (47)$$

To make the spatial and specification effects explicit, for a sourcing radius $R > 0$ define the grade-resolved compatible local supply vector

$$S_{jr}(R) := \int_{\mathbb{E}_L} \mathbf{1}\{d_D(\ell, \ell_j) \leq R\} \mathbf{1}\{\mathbf{g}_j(\lambda) = r\} \tilde{\Lambda}_t(d\lambda), \quad \mathbf{S}_j(R) := (S_{j1}(R), \dots, S_{jn_j^G}(R)). \quad (48)$$

Let $S_j(R) := \sum_{r=1}^{n_j^G} S_{jr}(R)$. Let $\varpi_j^{\text{srch}}(\mathbf{S}_j(R))$ be a procurement/search-risk premium that is nonincreasing in compatible supply in the relevant grade coordinates. For a compatible log of grade $r = \mathbf{g}_j(\lambda)$, define the least non-stumpage delivered cost

$$c_j^*(\lambda) := \inf_{k \in \mathbf{K}_t} c_t(\lambda, j, k) \quad (49)$$

and the quantity-sensitive processor reservation net value

$$\bar{v}_j(\lambda, \mathbf{S}_j(R), \mathbf{M}_j^\eta) := \pi_{jr}^{\text{marg}}(M_{jr}^\eta) - c_j^*(\lambda) - \varpi_j^{\text{srch}}(\mathbf{S}_j(R)). \quad (50)$$

When $\pi_{jr}^{\text{marg}} = p_{jr}$ and the search-risk term vanishes, this reduces to the best compatible posted-value net of delivered cost. Replacing v_j in (47) by $\bar{v}_j$ gives

$$p_j^{\text{FG}}(\lambda, \mathbf{S}_j(R), \mathbf{M}_j^\eta) = o_j(\lambda) + \beta_{\text{barg}}(\bar{v}_j - o_j(\lambda)). \quad (51)$$

Proposition 7 (Information, local scarcity, and negotiated value). *Fix a compatible grade r , log state λ , and radius R . Suppose all terms are differentiable along a scalar information state z on a region where $\bar{v}_j > o_j$. Then*

$$\begin{aligned} \frac{dp_j^{\text{FG}}}{dz} = & (1 - \beta_{\text{barg}}) \frac{do_j}{dz} + \beta_{\text{barg}} \left[(\pi_{jr}^{\text{marg}})'(M_{jr}^\eta) \frac{dM_{jr}^\eta}{dz} - \frac{dc_j^*}{dz} \right. \\ & \left. - \nabla \varpi_j^{\text{srch}}(\mathbf{S}_j(R)) \cdot \frac{d\mathbf{S}_j(R)}{dz} \right]. \end{aligned} \quad (52)$$

Hence improved information raises negotiated value whenever the combined outside-option, delivered-cost, and search-risk gains dominate any reduction in marginal procurement value caused by quantity saturation.

Proof. Differentiate (51) and (50). If marginal willingness to pay is nonincreasing then $(\pi_{jr}^{\text{marg}})' \leq 0$ wherever differentiable, while coordinatewise nonincreasing search risk gives $\nabla \varpi_j^{\text{srch}} \leq 0$. The stated sign condition follows term by term. $\square$

Thus a mapped cluster of mature, high-quality trees close to a processor is valuable for mathematically distinct reasons: it can reduce delivered/search cost, increase feasible matched surplus, and, when the same material is compatible with alternative buyers, improve the seller-side outside option. If the mapping reveals previously unknown compatible stock without immediately increasing contracted quantity, the search-risk term can raise processor reservation value. If supply is already abundant relative to demand, diminishing marginal procurement value or monopsony can instead dominate. Negotiating leverage is therefore a property of the local compatibility graph, grade-resolved quantity state, and their shadow values, not of hectares alone.

5 Partial observation and Bayes-adaptive control

5.1 Latent state, observations, and belief state

The physical state is not observed exactly. Let

$$X_t = (\mu_t, \nu_t, \kappa_t, r_t, \zeta_t) \in \mathbf{X} \quad (53)$$

where $r_t \in \mathbf{R}$ is a discrete observation/coverage regime and ζ_t collects finite-dimensional auxiliary states such as sensing backlog, active spatial domain, and computational or contractual states. Let $\theta \in \Theta$ contain unknown structural parameters: growth and quality-transition coefficients, bucking/recovery parameters, operator-productivity coefficients, processor acceptance and price-response parameters, observation-error parameters, and marginal sensing/operating costs.

At decision time t , let

$$u_t = (a_t^{\text{obs}}, a_t^{\text{exp}}, a_t^{\text{match}}, a_t^{\text{contract}}, a_t^{\text{cov}}, \iota_t) \in \mathbf{U}, \quad (54)$$

where the continuous components control sensing/verification, designed experiments, exposure of feasible matches, contract or posted-price variables, and spatial coverage intensity; ι_t is a discrete regime impulse. This control vector is distinct from the biological management action a_t^G in (20). No particular institutional objective is imposed: the one-step loss below may represent negative matched surplus plus sensing/operating cost, a planner objective, or another specified criterion. The state and observation kernels are

$$X_{t+1} \sim P_\theta(\cdot \mid X_t, u_t), \quad Y_{t+1} \sim G_\theta(\cdot \mid X_t, u_t, X_{t+1}). \quad (55)$$

Observations may include plot measurements, processor intake/acceptance, grade outcomes, contractor productivity, delivered-cost components, contract responses, repeated procurement decisions, or realised matches.

Let $\mathcal{I}_t$ be the observation/control filtration. The correct information state is the joint posterior

$$\pi_t(dx, d\theta) := \mathbb{P}((X_t, \theta) \in dx \times d\theta \mid \mathcal{I}_t) \in \mathcal{P}_2(\mathbf{X} \times \Theta), \quad (56)$$

not merely a posterior over parameters conditional on a perfectly known physical map.

Proposition 8 (Belief-state sufficiency). *If (P_θ, G_θ) are Borel Markov kernels and admissible actions are measurable functions of the information history, then π_t is a controlled Markov information state. Any finite-horizon history-dependent policy has an equivalent policy measurable with respect to π_t .*

Proof. The posterior predictive law of $(X_{t+1}, Y_{t+1}, \theta)$ is obtained by integrating the kernels against π_t . Bayes' rule maps (π_t, u_t, Y_{t+1}) to π_{t+1} ; hence future conditional laws depend on the past only through (π_t, u_t) . $\square$

5.2 Bellman equation and the Wasserstein HJB layer

Let $L(x, \theta, u)$ be one-step loss, let $\bar{L}(\pi, u) = \int L(x, \theta, u) \pi(dx, d\theta)$, and fix a discount factor $\gamma \in (0, 1]$. If $\mathcal{K}(d\pi' \mid \pi, u)$ is the Bayesian filtering kernel, let $\mathbf{U}(\pi) \subseteq \mathbf{U}$ denote the admissible action correspondence at belief π . The finite-horizon belief-state recursion is

$$V_t(\pi) = \inf_{u \in \mathbf{U}(\pi)} \left\{ \bar{L}(\pi, u) + \gamma \int V_{t+1}(\pi') \mathcal{K}(d\pi' \mid \pi, u) \right\}. \quad (57)$$

Proposition 9 (Attainment in the belief-state Bellman step). *Fix t . Suppose $\mathcal{U}(\pi)$ is nonempty compact with closed graph, $\bar{L}(\pi, u)$ is lower semicontinuous, $\mathcal{K}$ is weakly continuous in (π, u) , and V_{t+1} is lower semicontinuous and bounded below. Then the Bellman objective in (57) is lower semicontinuous in u and its infimum is attained for every π .*

Proof. Weak continuity of $\mathcal{K}$ and lower semicontinuity of V_{t+1} , together with the lower bound, imply lower semicontinuity of the continuation integral by the Portmanteau theorem. Adding $\bar{L}$ preserves lower semicontinuity. Compactness of $\mathcal{U}(\pi)$ gives attainment. $\square$

The continuous biological layer has its own infinite-dimensional dynamic-programming geometry. Suppose that $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is the conditional tree law, that a biological control $a^G \in \mathbf{A}_G$ enters the coefficients of (20), and that $V_G(t, \mu)$ is smooth in the Lions sense. For a fixed mark class ς and deterministic age path, write $b_G(t, x, \mu, a^G) := b_G(x, \mu, a_t, \varsigma, a^G)$, and analogously for σ_G and σ_G^0 . Let $F_G(t, \mu, a^G)$ be the running biological objective and $G_T(\mu)$ the terminal objective. Formally,

$$\begin{aligned} 0 = & \partial_t V_G(t, \mu) + \inf_{a^G \in \mathbf{A}_G} \left\{ F_G(t, \mu, a^G) + \int \partial_\mu V_G(t, \mu)(x) \cdot b_G(t, x, \mu, a^G) \mu(dx) \right. \\ & + \frac{1}{2} \int \text{Tr}[(\sigma_G \sigma_G^\top + \sigma_G^0 (\sigma_G^0)^\top)(t, x, \mu, a^G) \partial_x \partial_\mu V_G(t, \mu)(x)] \mu(dx) \\ & \left. + \frac{1}{2} \iint \text{Tr}[\sigma_G^0(t, x, \mu, a^G) \sigma_G^0(t, y, \mu, a^G)^\top \partial_{\mu\mu}^2 V_G(t, \mu)(x, y)] \mu(dx) \mu(dy) \right\}, \end{aligned} \quad (58)$$

with $V_G(T, \mu) = G_T(\mu)$. Fixed site/species/management marks have been suppressed in (58) for readability. The second Lions-derivative term is generated by common noise. Under suitable regularity assumptions, this DPP/HJB structure is rigorous; see Pham and Wei (2016); Djete et al. (2019). Intrinsic viscosity formulations on Wasserstein space and fully second-order equations generated by common noise are developed in Burzoni et al. (2019); Soner and Yan (2022); Bayraktar et al. (2025). The equation is used here to expose the geometry and derivative structure, not to assert that the full partially observed market-control problem has a closed-form HJB.

5.3 Risk and distributional robustness

For dynamic risk, use a one-step conditional risk map $\rho_{\pi, u}$. A nested expectation–CVaR recursion is

$$V_t^p(\pi) = \inf_{u \in \mathcal{U}(\pi)} \left\{ \bar{L}(\pi, u) + \gamma \rho_{\pi, u}(V_{t+1}^p(\pi_{t+1})) \right\}, \quad (59)$$

where

$$\rho_{\pi, u}(Z) = (1 - \lambda_R) \mathbb{E}[Z \mid \pi, u] + \lambda_R \text{CVaR}_{\alpha_R}(Z \mid \pi, u), \quad \lambda_R \in [0, 1], \quad \alpha_R \in (0, 1). \quad (60)$$

Nested conditional risk measures preserve time consistency in the dynamic setting (Yu and Shen, 2023). For one-period loss,

$$\text{CVaR}_{\alpha_R}(Z) = \min_{z \in \mathbb{R}} \left\{ z + \frac{1}{1 - \alpha_R} \mathbb{E}[(Z - z)_+] \right\}, \quad (61)$$

whose epigraph is convex whenever Z is convex in the decision variables.

Posterior uncertainty does not protect against model misspecification. Let $\hat{P}(\cdot \mid \pi, u)$ be a posterior predictive law on the next belief state, with W_2 induced by a compatible ground metric on the belief space, and define the ambiguity set

$$\mathfrak{P}_{\text{DR}}(\pi, u) = \{\mathbb{Q} : W_2(\mathbb{Q}, \hat{P}(\cdot \mid \pi, u)) \leq \varepsilon_{\text{DR}}(\pi, u)\}. \quad (62)$$

The robust Bellman operator replaces the expectation in (57) by the worst-case expectation over $\mathfrak{P}_{\text{DR}}$. Wasserstein distributionally robust programs can admit finite convex reformulations under useful structural assumptions (Esfahani and Kuhn, 2015); distributionally robust chance constraints are also consistent under suitable sampling assumptions (Cherukuri and Hota, 2020).

5.4 Dual control and experimental design

Because u_t changes the observation distribution, actions alter both current payoff and future posterior uncertainty. Let $V_{t+1}(\pi_t^{\text{freeze}})$ denote continuation value if the next observation is ignored. Under a loss convention, define the endogenous learning value

$$\mathcal{D}_t(\pi, u) = \mathbb{E} \left[V_{t+1}(\pi_{t+1}^{\text{freeze}}) - V_{t+1}(\pi_{t+1}) \mid \pi_t = \pi, u_t = u \right]. \quad (63)$$

If the controller may ignore information, $\mathcal{D}_t(\pi, u) \geq 0$ whenever the observation has no additional cost beyond that already charged in L . This is the dual-control effect: a price experiment, contractor-instrumentation protocol, or targeted inventory sample may have lower immediate contribution but higher continuation value because it changes the next action set through learning (Meijer and Rantzer, 2026; Li et al., 2023).

Let $e \in \mathbf{E}_{\text{exp}}$ denote an admissible experiment/instrumentation design. An information-theoretic design criterion is

$$\text{EIG}(e) = \mathbb{E}_{Y|e} [\text{KL}(\pi(\cdot \mid Y, e) \parallel \pi(\cdot))], \quad (64)$$

while a decision-focused expected value of sample information under the loss convention can be written

$$\text{EVSI}(e) := \inf_u \mathbb{E}[L \mid \pi] - \mathbb{E}_{Y|e} \left[\inf_u \mathbb{E}[L \mid Y, e, \pi] \right], \quad (65)$$

for the relevant decision horizon (or with L replaced by continuation value in the dynamic problem). EIG and EVSI need not select the same experiment; the latter is the more direct criterion when observations affect an irreversible regime transition. Bayesian experimental-design methods provide tractable approximations to these nested expectations (Tsilifis et al., 2015; Foster et al., 2019; Rainforth et al., 2024).

5.5 Discrete regime switching as impulse control

Let $\mathbf{R} = \{r_1, r_2, r_3\}$ be a finite ordered set of observation/coverage regimes and let $\mathbf{X}_r \subseteq \mathbf{X}$ denote the physical state space conditional on regime $r \in \mathbf{R}$. For an admissible transition $r \rightarrow r^+$, let $\mathcal{K}_{r,r^+} \geq 0$ be its impulse cost and let

$$\mathcal{R}_{r \rightarrow r^+} : \mathcal{P}_2(\mathbf{X}_r \times \Theta) \longrightarrow \mathcal{P}_2(\mathbf{X}_{r^+} \times \Theta) \quad (66)$$

be a measurable posterior embedding/prediction operator. Let $\mathcal{C}_r$ denote the Bellman continuation operator when the regime is held fixed and let V_r^{stop} be a specified terminal/abandonment value. The quasi-variational recursion is

$$V_r(\pi) = \min \left\{ (\mathcal{C}_r V_r)(\pi), \mathcal{K}_{r,r^+} + V_{r^+}(\mathcal{R}_{r \rightarrow r^+} \pi), V_r^{\text{stop}}(\pi) \right\}. \quad (67)$$

The value of waiting is therefore an optimal-stopping object generated by information and irreversibility: a transition is selected only when the posterior value of the enlarged feasible state/action region exceeds both within-regime continuation and the impulse cost.

6 A coarse finite-dimensional projection of the structural model

The preceding model is the structural object. For current data, a lower-dimensional projection is required. Let H denote covered hectares, N_A paying asset accounts, N_P paying processors, and let M_j be matched/visible processor tonnes from (43). Define operating surplus

$$\Pi = R_A + R_S - C. \quad (68)$$

6.1 Coarse asset functional and adoption response

For asset account i , let A_i^{ha} be hectares, S_i annual managed forestry expenditure per hectare, V_i^{asset} standing asset value per hectare, δ_i^{op} the proportional operational improvement enabled by information, and λ_i^{loss} avoidable annual loss/risk. A value envelope is

$$\mathcal{V}_{A,i} = A_i^{\text{ha}} (S_i \delta_i^{\text{op}} + V_i^{\text{asset}} \lambda_i^{\text{loss}}). \quad (69)$$

The current reduced model uses a monetisation rate m_A against the managed-spend base of paying accounts,

$$R_A = m_A B_A, \quad B_A = \sum_i C_i^A A_i^{\text{ha}} S_i, \quad (70)$$

where $C_i^A \in \{0, 1\}$ is adoption.

For a homogeneous segment, let adoption at price p be

$$\mathfrak{a}(p) = \frac{1}{1 + \exp(\eta_A p - \alpha_A)}, \quad \eta_A > 0, \quad (71)$$

and revenue $R(p) = Bp \mathfrak{a}(p)$ for $B > 0$.

Proposition 10 (Strict log-concavity of one-segment pricing). *For $p > 0$, $R(p) = Bp/(1 + e^{\eta_A p - \alpha_A})$ is strictly log-concave and therefore has a unique maximizer. The interior optimizer is*

$$p^* = \frac{1 + \text{LambertW}(e^{\alpha_A - 1})}{\eta_A}, \quad (72)$$

where LambertW denotes the principal real branch of the Lambert function.

Proof.

$$\frac{d^2}{dp^2} \log R(p) = -\frac{1}{p^2} - \eta_A^2 \mathfrak{a}(p)(1 - \mathfrak{a}(p)) < 0.$$

The first-order condition is $\eta_A p(1 - \mathfrak{a}(p)) = 1$. Setting $y = \eta_A p - 1$ gives $ye^y = e^{\alpha_A - 1}$, hence (72). $\square$

This result is local to a homogeneous segment. Mixtures of customer segments, discrete products, cross-side network effects, and dynamic learning can destroy global concavity.

6.2 Coarse matched-supply functional

Let f_j be the fee per tonne of processor-relevant supply made commercially visible or matchable. Using the structural matching output,

$$R_S = \sum_j C_j^S f_j M_j, \quad (73)$$

where $C_j^S \in \{0, 1\}$ is processor adoption. The previous scalar coverage discount is thus interpreted as an approximation to the matched-flow operator $M_j(\mu, \nu, \kappa)$, not as a primitive.

6.3 Structural cost and operating leverage

Use the cost function

$$C(H, N_A, N_P, n_V, W_D, \mathcal{G}_{\text{act}}) = F + \sum_{g \in \mathcal{G}_{\text{act}}} F_g + a_H H^{\beta_H} + b_A N_A + b_P N_P + c_V n_V + c_D W_D + \varepsilon_C, \quad (74)$$

where $\mathcal{G}_{\text{act}}$ is the active geography set, F and F_g are common and geography-specific fixed costs, $a_H, b_A, b_P, c_V, c_D \geq 0$ are marginal/intensity coefficients, $\beta_H > 0$ is the variable coverage exponent, n_V is verification count, W_D is data/compute workload, and ε_C is a zero-centered residual cost shock in the stochastic version. The key hypothesis is that verification is selected by posterior uncertainty and decision value rather than imposed as a census of all covered hectares.

Proposition 11 (Fixed-cost dilution). *If $C(H) = F + a_H H^{\beta_H}$ with $F, a_H, \beta_H > 0$, then*

$$\frac{d \log C}{d \log H} = \beta_H \frac{a_H H^{\beta_H}}{F + a_H H^{\beta_H}} < \beta_H. \quad (75)$$

Hence an observed decline in average cost per hectare does not identify sublinear marginal coverage cost.

Let $R_H := R_A(H) + R_S(H)$, define the local revenue elasticity $\beta_R(H) := d \log \mathbb{E}[R_H] / d \log H$, and let $\beta_C(H)$ denote the corresponding local elasticity of expected recurring cost. Positive operating leverage requires

$$\beta_R(H) > \beta_C(H). \quad (76)$$

The current three cost anchors are compatible with a large fixed component and a variable exponent near one; section 8 reports the fit.

6.4 Break-even, risk, and robustness

At fixed scale, the deterministic monetisation frontier is

$$m_A B_A + \sum_j C_j^S f_j M_j = C. \quad (77)$$

If a common reduced-order supply fee f_S is imposed, define the adopted matched-tonnage base $B_S := \sum_j C_j^S M_j$. For two scalar monetisation rates (m_A, f_S) and fixed bases (B_A, B_S) , the feasible break-even half-space

$$\{(m_A, f_S) \in \mathbb{R}_+^2 : m_A B_A + f_S B_S \geq C\} \quad (78)$$

is convex.

Let $R_H > 0$ and $C_H > 0$ be random recurring revenue and cost at scale H , and let $\tau \in (0, 1)$. Define the multiplicative revenue factor required for probability τ of nonnegative operating surplus by

$$M_H^\tau := \inf\{k > 0 : \mathbb{P}(kR_H - C_H \geq 0) \geq \tau\}. \quad (79)$$

Define the left quantile $Q_\tau(Z) := \inf\{z \in \mathbb{R} : \mathbb{P}(Z \leq z) \geq \tau\}$. Then

$$M_H^\tau = Q_\tau\left(\frac{C_H}{R_H}\right). \quad (80)$$

This identity follows directly from $R_H > 0$ and the definition of the generalized quantile; no continuity assumption on the ratio distribution is required.

7 Identification and numerical approximation

7.1 What must be identified

The structural parameter vector contains at least four blocks:

$$\theta = (\theta_G, \theta_B, \theta_M, \theta_C), \quad (81)$$

where θ_G governs growth/quality dynamics, θ_B harvest/bucking and log-yield maps, θ_M processor acceptance, adoption, matching and bargaining, and θ_C service/verification cost. A useful observation model is

$$Y = \mathfrak{F}_{\text{obs}}(X, u, \theta) + \delta_{\text{model}} + \varepsilon, \quad (82)$$

where $\mathfrak{F}_{\text{obs}}$ is the structural observation map, δ_{model} is structural discrepancy, and ε is observation noise. Without discrepancy, posterior concentration can become false precision. Bayesian calibration with a discrepancy process is standard in computer-model calibration (Kennedy and O’Hagan, 2001); identifiability requires experiments that separate parameter effects from discrepancy (Wu et al., 2018).

The data hierarchy is correspondingly explicit:

1. tree/plot data identify growth, size distribution, mortality, and quality transitions;
2. harvest/bucking records identify the map from standing trees to log distributions;
3. contractor records identify productivity and delivered-cost kernels;
4. processor intake records identify grade acceptance and realised price schedules;
5. repeated bids and alternative buyers identify bargaining/outside-option structure;
6. platform conversion and renewal identify willingness to pay and adoption elasticity.

7.2 Simulation architecture

A structurally faithful numerical experiment is a nested stochastic program over a measure-valued state, not a single Monte Carlo draw over scalar parameters. For posterior particle $s = 1, \dots, N_\theta$, one numerical path consists of:

1. draw $\theta^{(s)} = (\theta_G, \theta_B, \theta_M, \theta_C)$ from the current posterior and a common environmental path;
2. propagate a particle approximation of the conditional growth law (20), using the same common-noise realization across trees in a site class and idiosyncratic noise at tree level;
3. estimate mortality/harvest events, then solve or sample the bucking map to obtain a random marked log measure with dimensions and quality;

4. transform log volume to mass through (3) and construct $\tilde{\Lambda}_t$;
5. build the sparse feasible hypergraph (λ, j, k) induced by processor grade sets, sourcing distance and operator feasibility;
6. solve the exact linear program (38) or the entropy-regularized problem (41), retaining primal flows and dual potentials;
7. generate observations under the selected sensing/contracting action and update the joint state-parameter posterior;
8. evaluate dynamic risk, continuation value and any scale impulse under the candidate policy.

The physical discretization should preserve the distributional quantities that enter processor acceptance. A useful hierarchy is: individual-tree particles for DBH/height/form uncertainty; adaptive log bins in $(d_s, d_\ell, L_L, q_L, w)$; processor/grade nodes; and operator-capacity nodes. Binning only by tonnes is insufficient because price is discontinuous at specification boundaries. Near such boundaries, adaptive refinement or importance sampling is preferable to a uniform grid.

The McKean–Vlasov layer may be advanced in time by Euler–Maruyama/Milstein particles or by a learned conditional transition density. The empirical measure $N^{-1} \sum_i \delta_{\Xi_i}$ approximates the conditional law within a mark/site class; common-noise coupling must be preserved rather than independently resampled across trees. Size-structured harvesting models provide a complementary PDE/transport limit when individual diffusion is less useful (Wang and Yu, 2026), while empirical diameter-distribution models can be used as static or initialization diagnostics (Teimouri et al., 2019).

The allocation layer is conditionally convex. Its exact LP provides a benchmark; the entropy-transport approximation enables Sinkhorn-type scaling (Chizat et al., 2016). Regularization bias should be estimated by solving a subset of scenarios at decreasing ε and comparing both primal value and dual potentials. Those duals act as a sensitivity oracle: if an outer sensing or coverage decision changes (S, D, O) affinely, the dual variables provide supergradients of the concave matching value and support Benders/cutting-plane decompositions. Discrete grade, minimum-lot, harvest and regime choices remain outside this convex core.

The outer control problem can then be approximated by posterior scenario trees, sequential Monte Carlo with rollout, stochastic model-predictive control, or fitted value functions on a low-dimensional belief embedding. Reduced-order coordinates should be chosen from quantities with direct decision meaning—for example local compatible supply $S_j(R)$, matched-demand fractions χ_j , posterior moments of growth/quality, and selected dual potentials—rather than from raw hectares. This is a natural place to exploit state-space reduction without collapsing the tree/log specification structure.

For the current coarse benchmark, sample-average approximation is used only as a prior-predictive diagnostic (Kleywegt et al., 2002). Correlated uncertainty is assessed with Shapley effects rather than an independence-dependent Sobol decomposition (Iooss and Prieur, 2017). Numerical uncertainty should be decomposed into at least five components: particle/discretization error in the biological law, posterior/filter approximation error, bucking/taper-model error, transport regularization/scenario error, and structural model discrepancy. Reporting a single Monte Carlo interval that mixes these objects would obscure where additional data or computation is valuable.

8 East African numerical calibration

Only at this point do we impose current East African evidence. Internal analysis estimates approximately 338,555 planted hectares in Uganda and 588,000 hectares in Tanzania, approximately 0.93 million hectares combined (Gatsby Africa, 2026a). A processor intelligence base identifies 37

Chinese-owned processors, approximately 5 million m³/year of installed capacity, approximately 3 million m³/year of current use, and typical utilisation around 50–60% (Gatsby Africa, 2026b). These quantities initialize one numerical calibration; they are not assumptions of the general model. Reported processor requirements occur in both cubic metres and tonnes; the structural model therefore converts only through the state-dependent density/volume map in (3), never through a universal tonnes-per-cubic-metre constant.

The available evidence already supports the need for the richer specification layer. Processor records include heterogeneous raw-material descriptions, diameter/quality preferences, catchment radii, current and required procurement volumes, utilisation, and sourcing constraints; the existing roundwood model separately represents mensuration, felling, extraction, loading and haulage primitives. In the internal processor-supply workbook, for example, some processors require high-grade consistent roundwood while others explicitly accept smaller-dimension or lower-quality fast-growing material, so a single regional price-per-tonne process is structurally misspecified (Gatsby Africa, 2026d). The public repository also contains grade/specification data structures but labels several records as dummy functional test data; those records are therefore excluded from empirical calibration (EA Forests, 2026).

The same repository contains trial-site records indexed by age, genetic entry and site. They report DBH, height, stem-form score, survival and volume-proxy summaries or distributions. These variables can initialize a conditional growth/quality prior, but they do not identify a mature-tree transition kernel or a processor-grade recovery kernel. Longitudinal remeasurement and harvest/bucking observations are therefore required before the structural simulator can be treated as calibrated rather than prior-predictive (EA Forests, 2026).

8.1 Asset structure and current coarse projection

The Uganda CFR register used in the current calibration contains approximately 94,820 recorded planted hectares across 3,294 positive planted portfolios. The planted-area Gini coefficient is approximately 0.803; the largest portfolio represents about 10.4% of recorded planted hectares and the top five about 29.3% (Gatsby Africa, 2026c). Thus area scale cannot be converted to account count without the empirical portfolio distribution.

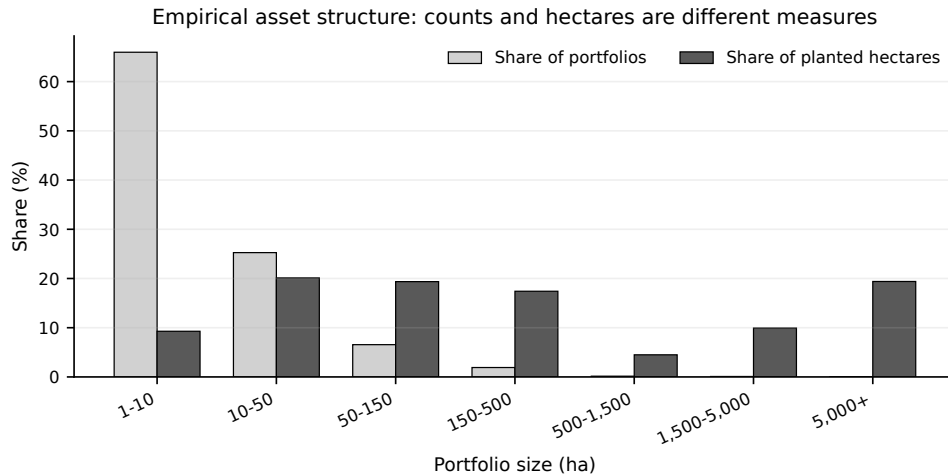


Figure 1: Empirical Uganda CFR portfolio structure used in the reduced-order initialization.

The current calibration does not yet contain a region-wide joint tree-level size–quality distribution. It therefore replaces the structural mapping $\mu_t \mapsto \tilde{\Lambda}_t \mapsto \Gamma_t^*$ with a conservative effective commercially available flow prior and processor-coverage priors. These are placeholders to be replaced as tree, contractor, and intake data arrive.

8.2 Current prior-predictive calibration

The reduced model uses observed grower structure, processor/catchment evidence, value-derived monetisation priors, and a correlated latent market state. The reported outputs are from 50,000 joint prior-predictive draws. This count controls numerical Monte Carlo error conditional on the model; it does not validate the structural priors or resolve model discrepancy.

Table 2: Current reduced-order prior-predictive scale results. USD figures are thousands/year.

	100k ha	500k ha	1m ha
Revenue P20	49	326	650
Revenue P50	78	451	850
Revenue P80	115	613	1,090
Base OPEX P50	219	424	669
Robust OPEX P50	350	617	875
Joint operating surplus P20	-197	-125	-66
Joint operating surplus P50	-145	21	171
Joint operating surplus P80	-97	191	436
$\mathbb{P}(\Pi > 0)$	0.01	0.55	0.72
Paying-ha penetration	0.25	0.26	0.262
Asset fee (USD/paying ha)	2.16	2.17	2.18
Supply fee (USD/visible t)	0.253	0.259	0.260

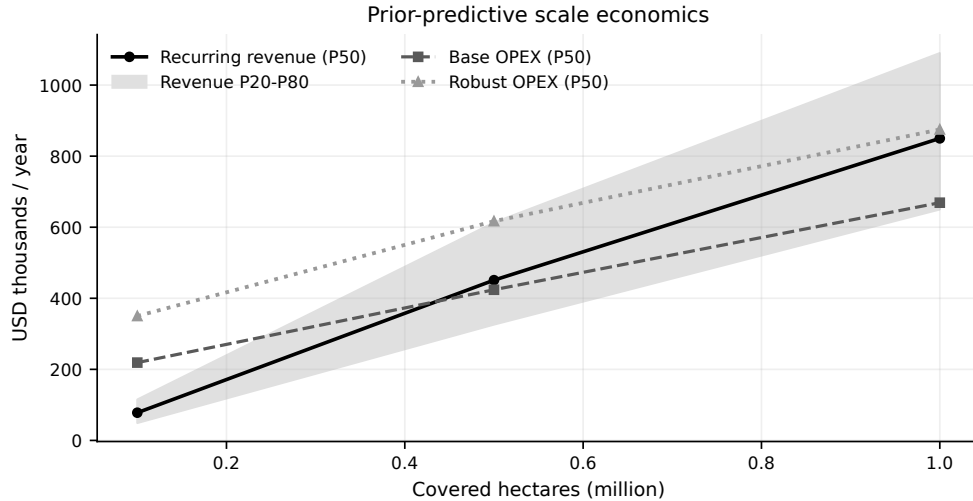


Figure 2: Reduced-order recurring revenue and OPEX distributions across the three coverage regimes.

The current base median cost anchors are approximately \$219k, \$424k, and \$669k at 100k, 500k, and one million covered hectares. A naive pure power fit yields an apparent total-cost exponent

near 0.47. The same three points are exactly fit by

$$C(H) = 162.8 + 56.2 \left(\frac{H}{100,000} \right)^{0.955} \quad (\text{USD thousands/year}), \quad (83)$$

which is consistent with large fixed cost plus nearly linear variable coverage cost.

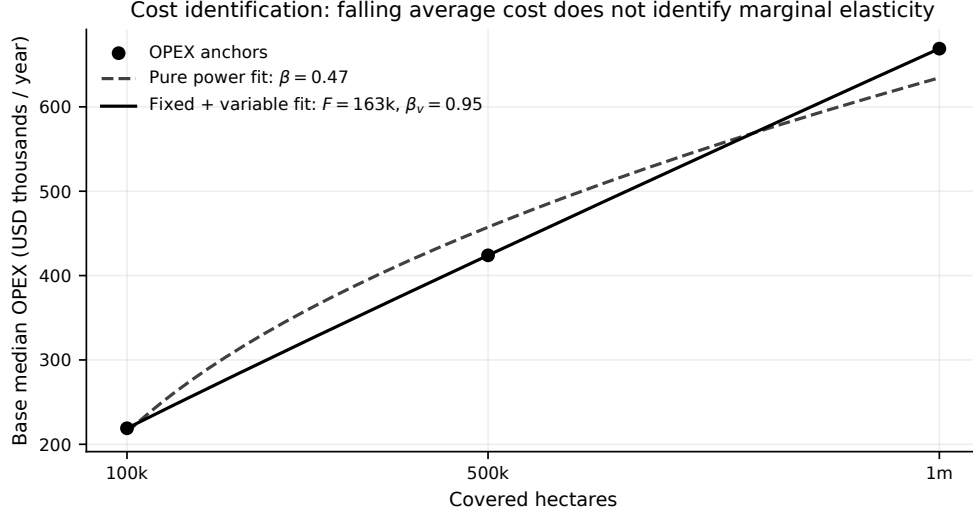


Figure 3: Three cost anchors do not identify strongly sublinear marginal cost; fixed-cost dilution can generate the same average-cost pattern.

8.3 Break-even and robustness

Holding the other revenue side fixed at its current median, one-lever deterministic break-even thresholds are:

Table 3: One-lever reduced-order break-even thresholds.

Scale	Cost mode	Asset monetisation (% managed spend)	Supply fee (USD/t)	Paying-ha penetration (%)
100k	Base	2.23	2.04	90.3
100k	Robust	3.73	3.70	150.9
500k	Base	0.56	0.21	23.5
500k	Robust	0.98	0.54	41.3
1m	Base	0.42	0.08	17.9
1m	Robust	0.65	0.29	27.3

The estimated 80%-probability robustness multipliers in (79) are

$$M_{100k}^{0.8} \approx 4.68, \quad M_{500k}^{0.8} \approx 1.37, \quad M_{1m}^{0.8} \approx 1.10. \quad (84)$$

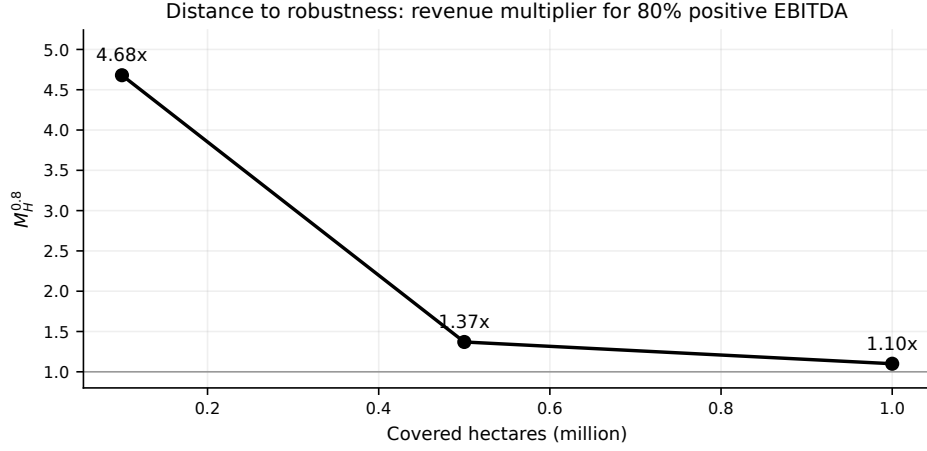


Figure 4: Reduced-order multiplier required to reach 80% probability of nonnegative recurring operating surplus.

These results are not evidence that hectares cause profitability. They state how far the current reduced-order distributions lie from a risk threshold under their assumed mapping from hectares to commercially useful coverage. The structural model identifies the missing empirical task: replace that mapping with the estimated tree-growth, bucking, matching, and service-cost operators.

8.4 What is and is not identified

The current evidence supports the physical scale of the market, heterogeneous processor demand, portfolio concentration, and several operating-cost primitives. It does not yet identify a region-wide joint distribution of tree diameter, length-recoverable volume, quality, moisture/density, harvest timing, and processor acceptance. Nor does it identify a stable bargaining function such as (47). Therefore the tree/log/matching layer should be treated as a model to be estimated, not as a source of invented precision.

This is also the strongest reason to work directly with contractors, operators, and processor intake data. Those observations identify K_B , $c(\lambda, j; k)$, realised grade maps, density conversions, acceptance rates, and the outside-option graph. Once these objects are estimated, the model can value a mapped cluster of mature trees near a processor in the correct unit: expected matched surplus and its dual shadow value, rather than hectares alone.

9 Conclusion

The natural state of a forestry market is measure-valued. Tree growth moves mass through size-quality space; harvest removes mass from the standing-tree measure and creates a log measure; processor specifications define admissible regions of log space; spatial operating costs define the edge weights of a supply-demand network; and market allocation is a partial transport problem on those measures. Because the state itself is only partially observed, the sufficient decision variable is a posterior distribution over both the physical state and unknown structural parameters.

This formulation separates several questions that are otherwise easily conflated. Biological productivity is not commercial availability. Commercial availability is not processor compatibility. Processor compatibility is not delivered surplus. Delivered surplus is not negotiated seller price.

Physical coverage is not information value. Each arrow between these objects is an operator that can be estimated and stress-tested.

The present East African numbers are therefore best viewed as a reduced-order prior predictive projection of the structural model. They motivate scale experiments but do not eliminate the need to identify the underlying operators. The analytical advantage of the richer formulation is precisely that new inventory, contractor, and processor data have a defined destination in the model: they update growth laws, bucking kernels, matching costs, grade acceptance, dual supply values, bargaining outside options, or service-cost functions. This makes the model progressively less dependent on hectare-level proxies as the information system itself becomes more informative.

A Notation audit

Principal symbols and domains. Distinct alphabets are used for physical states, controls, beliefs, and reduced-order projections to avoid collisions.

Symbol	Definition
$\mathbb{R}_+$	Nonnegative reals $[0, \infty)$
$\mathbf{E}_T, \mathbf{E}_L, \mathbf{E}_P, \mathbf{E}_O$	Tree, log, processor, and operator state spaces
$d_s, d_\ell, L_L, q_L, w, m_L$	Small-/large-end log diameters, log length, quality vector, moisture/density state, and log mass
$\mu_t, \Lambda_t, \tilde{\Lambda}_t$	Standing-tree finite measure, log-count measure, and tonne-weighted log measure
ν_t, κ_t	Processor and operator empirical measures
n_P, n_O	Numbers of active processors and operators
σ_j^{spec}	Finite-dimensional processor specification/grade/price configuration
$\bar{\mu}_t$	Normalized full marked-tree law from the finite standing-tree measure
$\Xi_t, \hat{\mu}_t^G$	Representative continuous tree state and its conditional size–quality law
$V_G(t, \mu)$	Value function for the conditional-law biological control subproblem
$\varsigma, a_t^G, \mathbf{A}_G$	Fixed biological marks, biological/management control, and compact metric action space
λ_t^H, λ_t^M	Harvest and mortality hazards
$\mu_t^\varsigma, \mathcal{B}_t^\varsigma$	Finite standing-tree measure and recruitment-rate measure restricted to a fixed mark class
$\mathfrak{L}_t^{\varsigma, a^G}$	Time-dependent conditional McKean–Vlasov diffusion generator on that class
$\mathcal{B}_t, \beta_t^B, K_B$	Recruitment-rate measure, bucking rule, and harvested-tree-to-log kernel
$\mathbf{A}_B(\xi), \mathcal{J}_B, \mathbf{v}_t^L, c_B$	Feasible bucking-pattern set, bucking objective, downstream log value, and bucking cost
D_t, O_t	Processor-demand and operator-capacity measures, in tonnes per decision period
$\mathfrak{R}_t, \varepsilon_\Gamma, \varepsilon_L, \varepsilon_D, \varepsilon_O$	Reference measure and entropy/marginal relaxation weights in unbalanced transport
Γ_t^*	Optimal log–processor–operator allocation measure
M_j, χ_j	Matched tonnes and matched-demand fraction for processor j
$n_j^G, \mathcal{S}_{jr}, \mathfrak{g}_j$	Number of processor grades, grade-acceptance sets, and highest-valued admissible grade map

continued on next page

Notation audit continued from previous page

Symbol	Definition
$a_{jr}^\eta, a_j^\eta, \bar{p}_j^\eta$	Grade-specific and total smoothed compatibility probabilities, and measurement-error-regularized expected posted value
$p_j^0(\lambda), \pi_{jr}^{\text{marg}}, \Phi_{jr}$	Posted grade price, grade-specific marginal procurement value, and cumulative grade-value function
$c_j^*, \varpi_j^{\text{srch}}, \bar{v}_j$	Least delivered cost, search-risk premium, and quantity-sensitive processor reservation net value
$M_{jr}^\eta(\Gamma), \mathbf{M}_j^\eta(\Gamma)$	Measurement-error-regularized expected grade tonnes and processor grade-quantity vector
$c_t(\lambda, j, k), s_t(\lambda, j, k)$	Delivered operating cost and compatible delivered surplus per tonne
$S_{jr}(R), \mathbf{S}_j(R)$	Grade-resolved local compatible supply and its vector within sourcing radius R
$\beta_{\text{barg}}, p_j^{\text{FG}}(\lambda, \mathbf{S}_j(R), \mathbf{M}_j^\eta)$	Seller bargaining weight and negotiated forest-gate net value in the stylized Nash split
ϕ, ψ, ω	Dual potentials for supply, demand, and operator capacity
$o_j(\lambda), v_j(\lambda)$	Alternative-buyer outside option and best net value with target processor j
X_t, r_t, ζ_t	Latent physical state, discrete observation/coverage regime $r_t \in \mathbf{R}$, and finite-dimensional auxiliary state
$\theta \in \Theta$	Structural parameter vector
π_t	Joint posterior law of (X_t, θ) given observations
γ	Dynamic-programming discount factor in $(0, 1]$
$e, \mathbf{E}_{\text{exp}}$	Experiment/instrumentation design and its admissible design space
$u_t, \mathbf{U}$	Sequential control and its continuous-plus-impulse action space in (54): sensing, experiment, matching, contract, spatial-coverage and regime controls
$\mathbf{R}, \mathbf{X}_r, \mathcal{C}_r, \mathcal{R}_{r \rightarrow r^+}, \mathcal{K}_{r, r^+}$	Finite regime set, regime-conditioned state space, continuation operator, posterior transition/embedding, and impulse cost
$\rho_{\pi, u}, \alpha_R, \lambda_R$	Conditional risk map, CVaR confidence level, and risk weight; ρ is unrelated to wood density ϱ
$\mathfrak{P}_{\text{DR}}, \varepsilon_{\text{DR}}$	Wasserstein ambiguity set and its radius in the distributionally robust Bellman problem
$\mathfrak{F}_{\text{obs}}, \delta_{\text{model}}, \varepsilon$	Structural observation map, model discrepancy, and observation noise
W_D	Data/compute workload entering the reduced structural cost function
H, N_A, N_P, n_V	Reduced-order covered hectares, paying asset/processor accounts, and verification count
m_A, f_j	Reduced-order asset monetisation rate and processor fee per matched tonne
B_A, B_S	Adopted asset managed-spend base and adopted matched-tonnage base
R_A, R_S, C, Π	Asset revenue, supply revenue, recurring cost, and operating surplus
$\tau, Q_\tau(Z), M_H^\tau$	Robustness probability level, left quantile of Z , and revenue multiplier required for probability τ of $\Pi \geq 0$

B Current reduced-order adoption priors

The current asset-adoption prior is size-dependent. At central price/value conditions, the base probability schedule by planted portfolio size is approximately

Portfolio (ha)	1–10	10–50	50–150	150–500	500–1,500	1,500–5,000	5,000+
Base adoption	0.1%	0.5%	5%	15%	35%	60%	80%

These are priors, not measured conversion rates. Regional processor adoption is likewise conditioned on engagement status and useful supply coverage and should be replaced by observed conversion and renewal data.

C Structural cost sensitivity

Using (83), changing the variable exponent while holding the 100k anchor fixed illustrates the sensitivity of scale economics:

Variable exponent	500k OPEX	1m OPEX
0.95	\$422k	\$664k
1.00	\$444k	\$725k
1.05	\$467k	\$793k
1.15	\$521k	\$957k
1.05 + \$100k geography block at 1m	\$467k	\$893k

D Reproducibility and data provenance

The numerical tables are generated from the canonical EA Forests analytical notebook and its scale-reference, stress-test, and frontier outputs. The empirical asset distribution is calculated from the Uganda CFR register. Aggregate processor and sector statistics are taken from internal 2025–2026 Gatsby Africa intelligence sources. The EA Forests codebase contains forestry cost recipes, harvesting/haulage primitives, processor locations, grade/specification schemas, and price schedules (EA Forests, 2026). Repository records explicitly labelled test/dummy data are excluded from empirical calibration.

A public preprint should either publish a de-identified calibration dataset sufficient to reproduce the reported numerical tables or mark the empirical case-study data as restricted. The theoretical sections are independent of those restricted data.

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